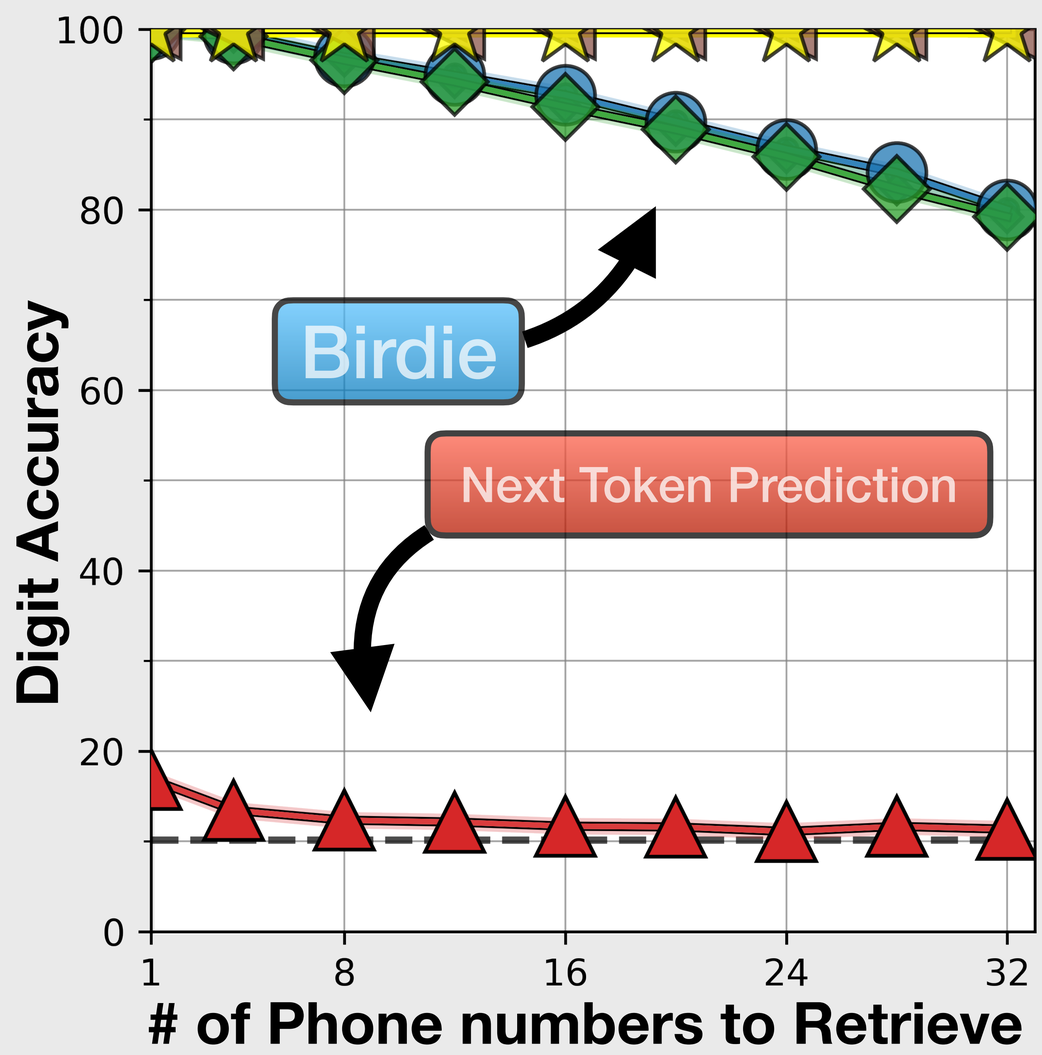
# Birdie

## Advancing State Space Models with Reward-Driven Objectives and Curricula

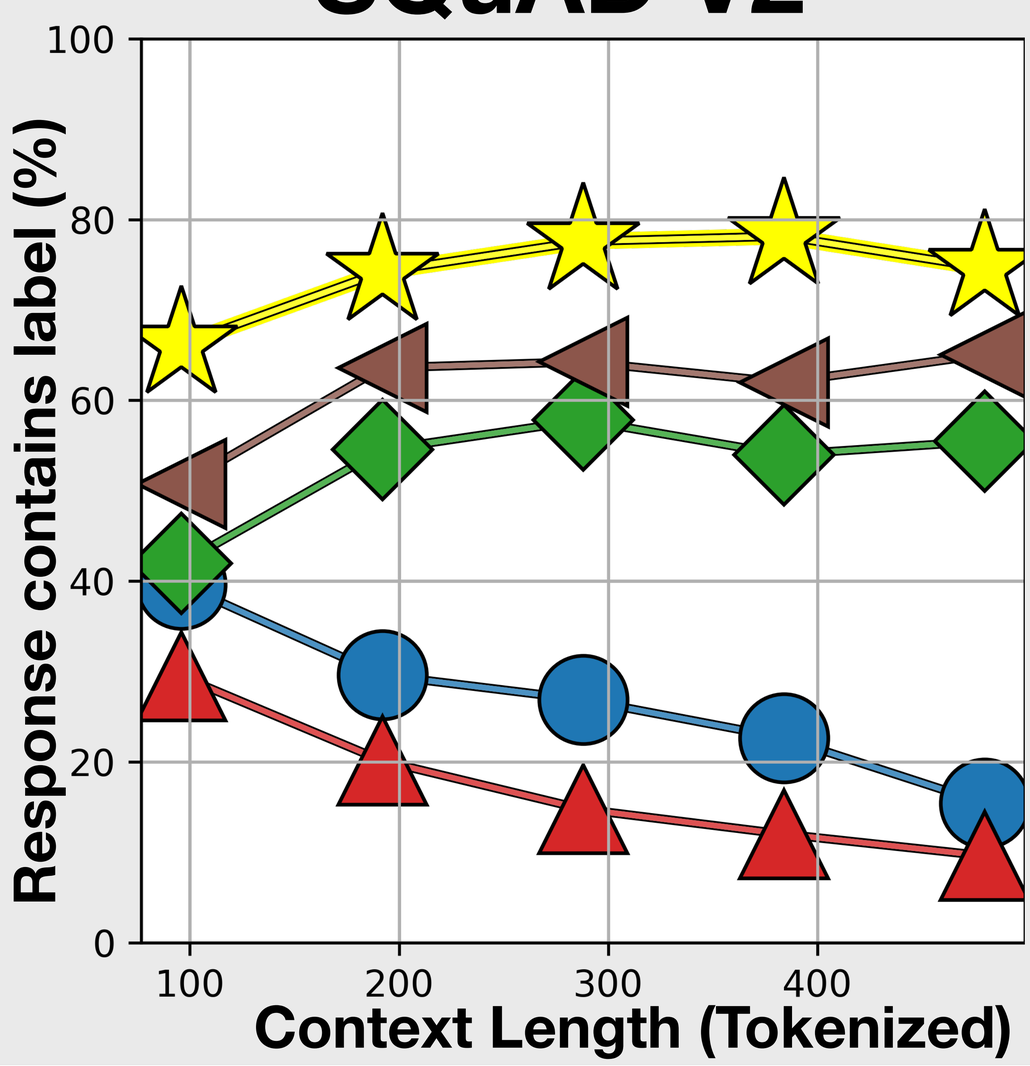
Sam Blouir, Jimmy Smith, Antonios Anastasopoulos, Amarda Shehu

## SSMs: Replace *Next Token Prediction* with *Birdie*

### Multi-Number Phonebook Task

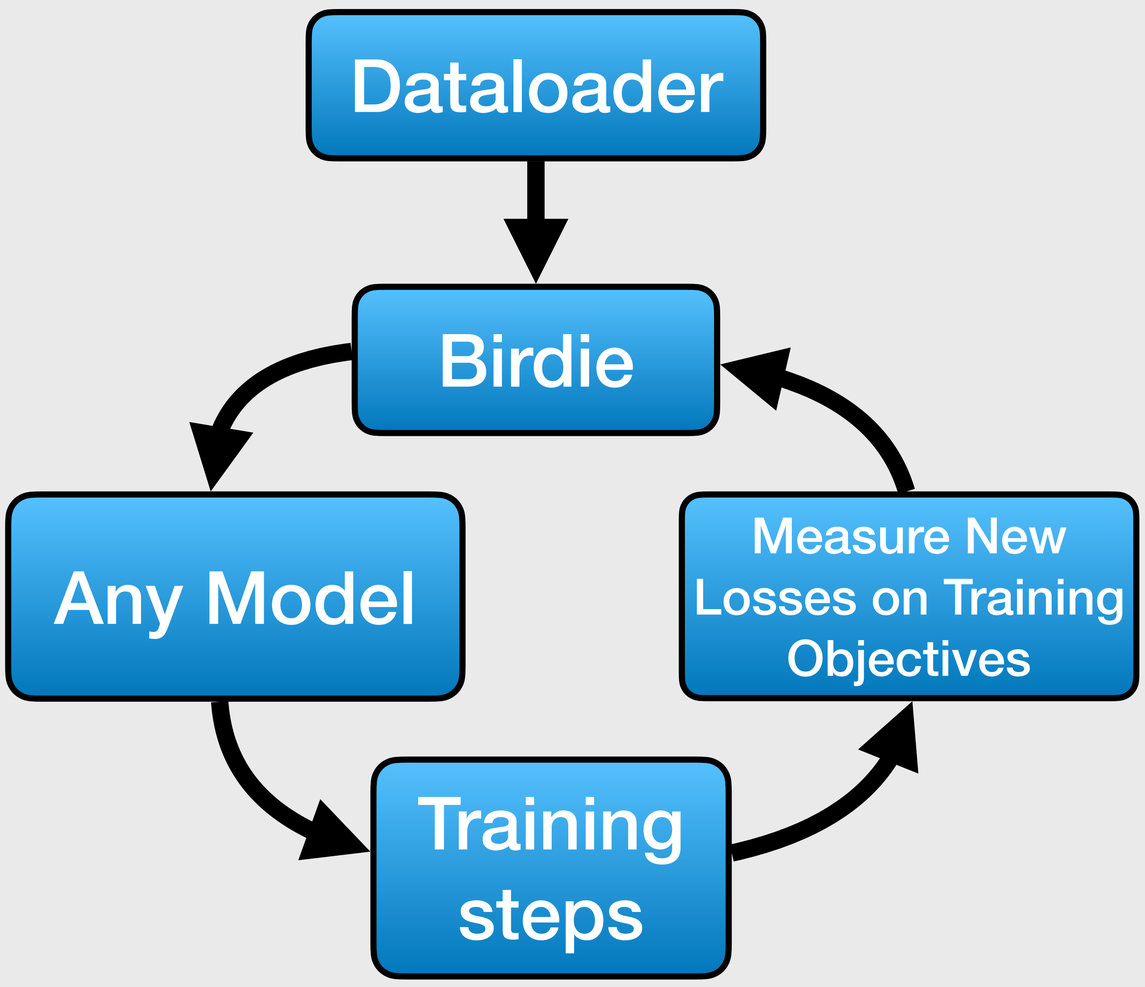
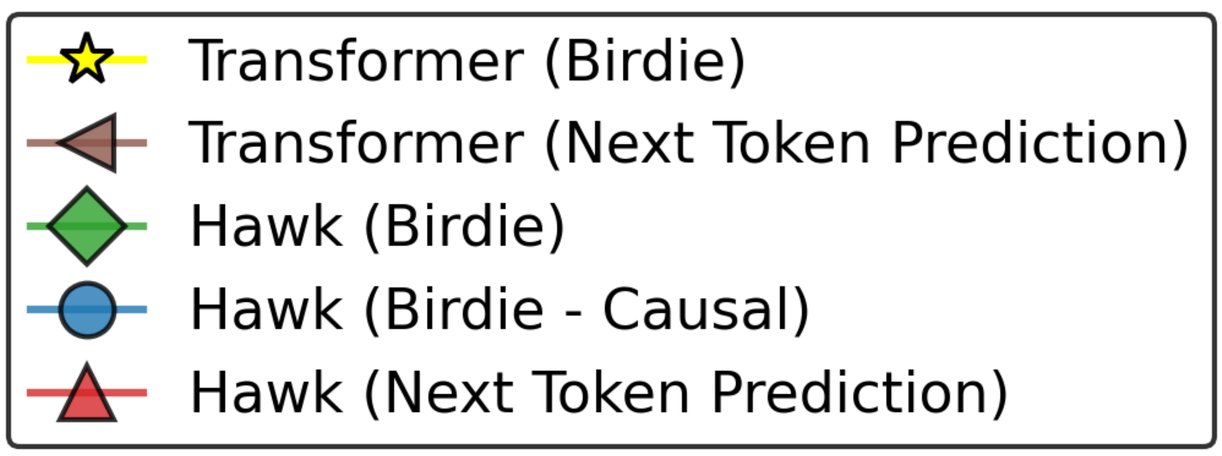



### SQuAD V2




## Birdie

1. Birdie trains models using **multiple self-supervised objectives**, including our novel Selective Copying objective introduced during pre-training.

2. We incorporate reinforcement learning into language model pre-training to estimate optimal parameters for these objectives. Birdie is capable of training models with diverse inductive biases, such as **Transformers** and **State Space Models (SSMs)**.

3. We evaluate performance on 21 standard tasks - including **ARC** and **MMLU** - as well as a new story infilling task. Results demonstrate that knowledge and modeling capabilities are preserved or enhanced, while context-recall abilities **significantly improve**.